



DECISION GOVERNANCE INFRASTRUCTURE

UNITED STATES PATENT AND TRADEMARK OFFICE
PROVISIONAL PATENT APPLICATION

TITLE OF INVENTION:

**ETHICAL AND QUANTUM-SECURE EXECUTION
FRAMEWORK WITH MULTI-TIER COMPLIANCE ENGINE
AND DUAL-LAYER POST-QUANTUM BIOMETRIC
AUTHORIZATION**

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This document constitutes a Provisional Patent Application filed pursuant to 35 U.S.C. § 111(b). It establishes a priority date and grants twelve (12) months from the filing date to submit a corresponding non-provisional application. This application has not been examined and does not confer patent rights independently. CONFIDENTIAL — NOT FOR DISTRIBUTION.

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PROVISIONAL APPLICATION FOR PATENT

Application Reference: OMNIX-PAT-2026-003

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Inventor: Harold Alberto Nunes Rodelo

SPECIFICATION

TITLE OF THE INVENTION

ETHICAL AND QUANTUM-SECURE EXECUTION FRAMEWORK FOR AUTOMATED FINANCIAL SYSTEMS:
INTEGRATED AUTOMATED SHARIA COMPLIANCE ENGINE WITH DIGITAL ASSET QUARANTINE AND
DUAL-LAYER TRANSACTION AUTHORIZATION USING VOICE BIOMETRICS AND POST-QUANTUM
CRYPTOGRAPHIC SIGNATURES

CROSS-REFERENCE TO RELATED APPLICATIONS

This application does not claim priority to any previously filed application. This provisional application is filed pursuant to 35 U.S.C. § 111(b) to establish a priority date. See also related provisional applications OMNIX-PAT-2026-001 (Governance Control Architecture) and OMNIX-PAT-2026-002 (Non-Markovian Signal Processing), filed concurrently herewith by the same inventor. The compliance engine and authorization framework described herein operate as integrated enforcement layers within the broader governance architecture described in OMNIX-PAT-2026-001.

STATEMENT REGARDING FEDERALLY SPONSORED RESEARCH OR DEVELOPMENT

Not applicable. This invention was made without Federal sponsorship.

TECHNICAL FIELD

The present invention relates to computer-implemented systems and methods for ethical compliance enforcement and quantum-resistant transaction authorization in automated financial decision systems. More specifically, the invention provides: (1) an Automated Sharia Compliance Engine that evaluates proposed digital asset transactions against Islamic financial jurisprudence criteria in real-time, prior to execution, and enforces compliance through a digital asset quarantine mechanism; and (2) a Dual-Layer Transaction Authorization system that combines voice biometric authentication with post-quantum cryptographic digital signature generation to provide transaction authorization that is resistant to both classical and quantum computing attacks. Both components are implemented as a unified execution framework operating as a pre-execution enforcement layer, connected through a common decision pipeline that evaluates compliance and identity in sequence before any transaction is permitted to proceed.

KEY DEFINITIONS

The following terms are used consistently throughout this specification and the appended claims:

"Sharia Compliance": Conformance of a financial transaction or instrument with the principles of Islamic financial jurisprudence (fiqh al-muamalat), including prohibition of riba (interest-based transactions), prohibition of gharar (excessive uncertainty or ambiguity in contract terms), and prohibition of maysir (speculation or gambling).

"Halal": Permissible under Islamic jurisprudence. A halal financial transaction is one that conforms to Sharia principles and is approved for execution.

"Haram": Impermissible under Islamic jurisprudence. A haram financial transaction is one that violates one or more Sharia principles and must be blocked from execution.

"Asset Quarantine": A computational enforcement mechanism that prevents any transaction involving a specific digital asset from being submitted for execution until the compliance classification of that asset has been verified or resolved. An asset quarantine is distinct from a transaction block: a transaction block applies to a specific proposed transaction, while an asset quarantine applies to all transactions involving the quarantined asset regardless of other transaction parameters.

"Voice Biometric Authentication": A computational process of verifying a user's identity by extracting and comparing acoustic features of a voice sample against a stored voice signature associated with an authorized user.

"Post-Quantum Cryptography (PQC)": Cryptographic algorithms designed to be resistant to attacks from quantum computers, as distinct from classical cryptographic algorithms (such as RSA and ECC) that are known to be vulnerable to quantum attacks using Shor's algorithm.

"Digital Signature": A cryptographic construct that provides non-repudiation, authentication, and integrity assurance for a digital message or document. In the context of this invention, a digital signature over a transaction payload provides cryptographic proof that the transaction was authorized by the identified user and that the transaction parameters have not been altered after authorization.

"Execution Boundary": The computational point at which a proposed transaction transitions from a recommendation or intent into an irreversible real-world action. The present invention enforces compliance and authorization at the execution boundary — after compliance and identity have been verified but before the transaction is transmitted to the execution environment.

"Quantum-Resistant": A property of a cryptographic algorithm indicating that it is not known to be efficiently solvable by a quantum computer using currently known quantum algorithms, including Shor's algorithm and Grover's algorithm.

"Dilithium-3 (ML-DSA-65)": The Module-Lattice-Based Digital Signature Standard at security level 3, as specified in NIST FIPS 204 (2024). This algorithm provides a security level equivalent to at least 128-bit classical security and is quantum-resistant.

"Kyber-768 (ML-KEM-768)": The Module-Lattice-Based Key Encapsulation Mechanism at security level 2, as specified in NIST FIPS 203 (2024). This algorithm provides quantum-resistant key encapsulation for establishing shared session keys.

BACKGROUND OF THE INVENTION

1. The Islamic Finance Market and the Compliance Technology Gap

The global Islamic finance industry represents a substantial and rapidly growing segment of international financial markets. As of 2024, total Islamic finance assets under management were estimated to exceed \$4 trillion USD, with significant concentrations in the Gulf Cooperation Council (GCC) states, Southeast Asia, and the United Kingdom. The core principle of Islamic finance is the prohibition of riba (interest), which necessitates fundamentally different structures for financial instruments compared to conventional finance.

The emergence of digital assets — including cryptocurrencies, tokenized securities, and stablecoins — has created a novel compliance challenge for Islamic finance participants. Unlike traditional financial instruments (equities, bonds, sukuk), digital assets present compliance classification challenges that are unique to the asset class:

(a) Novel instrument structures: Many digital assets do not correspond to any established category in classical Islamic finance jurisprudence. The absence of precedent requires case-by-case scholarly analysis rather than the application of established rulings.

(b) Dynamic compliance status: The compliance status of a digital asset may change over time as the underlying protocol, governance structure, or usage patterns evolve. A digital asset that was classified as halal at time T may become haram at time T+1 if its underlying protocol introduces interest-bearing mechanics or excessive speculation.

(c) Real-time execution requirements: Automated digital asset trading systems operate at speeds that make manual compliance review of individual transactions operationally impossible. Compliance enforcement must be embedded in the execution pipeline as a computational layer.

Existing approaches to Sharia compliance in financial institutions are primarily organizational and advisory: Sharia Supervisory Boards (SSBs) review financial products periodically and issue fatwas (Islamic legal opinions) that are then implemented through organizational policy. These mechanisms operate at time horizons of months, not milliseconds. No existing automated system provides real-time, transaction-level Sharia compliance enforcement with an asset quarantine mechanism for digital asset trading.

2. The Quantum Threat to Financial Transaction Security

Current financial transaction authorization systems rely predominantly on classical cryptographic algorithms for digital signatures and authentication, including RSA (Rivest-Shamir-Adleman) and ECDSA (Elliptic Curve Digital Signature Algorithm). These algorithms derive their security from the computational hardness of the integer factorization problem (RSA) and the elliptic curve discrete logarithm problem (ECDSA).

In 1994, Peter Shor demonstrated that a sufficiently powerful quantum computer could solve both of these problems in polynomial time, rendering RSA and ECDSA cryptographically insecure against a quantum adversary. While large-scale fault-tolerant quantum computers capable of running Shor's algorithm at cryptographically relevant scales do not yet exist as of the filing date of this application, the pace of quantum hardware development and the strategic implications of "harvest now, decrypt later" attacks — in which adversaries collect encrypted or signed data now for decryption or signature forgery after quantum computers become available — make the migration to post-quantum cryptographic standards an urgent infrastructure priority.

The United States National Institute of Standards and Technology (NIST) completed its post-quantum cryptography standardization process in 2024, publishing FIPS 203 (ML-KEM, key encapsulation) and FIPS 204 (ML-DSA, digital signatures) as the primary standards. As of the filing date of this application, no financial transaction authorization system is known to have combined voice biometric authentication with NIST FIPS 204-compliant post-quantum digital signatures for transaction authorization.

3. The Need for an Integrated Ethical and Quantum-Secure Execution Framework

The foregoing analysis reveals two distinct but complementary gaps in existing financial technology infrastructure:

- The absence of a real-time, transaction-level automated Sharia compliance enforcement system for digital assets with asset quarantine capability.
- The absence of a transaction authorization system combining biometric identity verification with post-quantum cryptographic signatures.

These two gaps share a common structural characteristic: both relate to the enforcement of pre-execution requirements at the execution boundary. Whether the pre-execution requirement is ethical (Sharia compliance) or identity-based (biometric authentication), the enforcement mechanism must operate at the same computational layer: after a transaction has been proposed and validated by the trading system but before it is transmitted to the execution environment.

The present invention provides a unified Ethical and Quantum-Secure Execution Framework that addresses both gaps through a single integrated pipeline, as described below.

SUMMARY OF THE INVENTION

The present invention provides a computer-implemented Ethical and Quantum-Secure Execution Framework comprising two integrated components connected by a unified execution pipeline:

Component 3A: Automated Sharia Compliance Engine with Digital Asset Quarantine

A real-time compliance enforcement system that evaluates proposed digital asset transactions against a Sharia compliance database prior to execution. The compliance database classifies digital assets according to a seven-tier compliance system derived from Islamic jurisprudence criteria including riba, gharar, and maysir prohibitions. The system includes an asset quarantine module that automatically restricts trading in quarantined assets until compliance status is resolved, and a periodic re-screening module that maintains classification currency over time.

Component 3B: Dual-Layer Transaction Authorization Using Voice Biometrics and Post-Quantum Cryptographic Signatures

A transaction authorization system that requires two sequential, independent verifications before a transaction is authorized for execution: (1) biometric identity verification through voice sample analysis and comparison against a stored voice signature; and (2) generation of a post-quantum digital signature over the transaction payload using Dilithium-3 (ML-DSA-65) as specified in NIST FIPS 204. Session establishment uses Kyber-768 (ML-KEM-768) as specified in NIST FIPS 203.

Unified Execution Pipeline — Execution Boundary Enforcement

Execution Boundary Principle: Both components of the Framework operate exclusively at the execution boundary — the precise computational point at which a proposed transaction would transition from an intent or recommendation into an irreversible real-world financial commitment. This placement is the defining architectural characteristic of the Framework. Compliance evaluation performed at the time of model training, at the time of system deployment, or through periodic organizational review does not constitute execution boundary enforcement. Identity verification performed through session login credentials or API key authentication does not constitute execution boundary enforcement at the transaction level. The present invention enforces both ethical compliance and identity authorization at the transaction level, at the moment of execution commitment, for every proposed transaction individually — not once per session, not once per day, but at the execution boundary of each individual proposed action.

The two components are connected through a unified execution pipeline that enforces the following sequence for every proposed transaction:

Step 1: Asset Compliance Check — The Sharia Compliance Engine evaluates the asset involved in the proposed transaction against the compliance database. If the asset is classified as haram or is quarantined, execution is blocked.

Step 2: Transaction Parameter Compliance Check — The Sharia Compliance Engine evaluates the transaction parameters (including leverage, interest components, and speculative characteristics) for compliance.

Step 3: Voice Biometric Authentication — Upon compliance clearance, the user's identity is verified through voice biometric authentication.

Step 4: Post-Quantum Signature Generation — Upon successful voice authentication, a post-quantum digital signature is generated over the complete transaction payload.

Step 5: Audit Receipt Generation — A complete audit record is generated, cryptographically sealed, and persisted.

Step 6: Execution Transmission — The signed transaction is transmitted to the execution environment.

This sequential pipeline ensures that no transaction is executed without both ethical compliance clearance and quantum-resistant identity authorization. Either failure independently blocks execution.

BRIEF DESCRIPTION OF THE DRAWINGS

FIG. 1 is a system architecture diagram illustrating the complete Ethical and Quantum-Secure Execution Framework, showing the relationship between the Sharia Compliance Engine, the Asset Quarantine Module, the Voice Biometric

Authentication Module, the PQC Signature Generation Module, and the Audit Receipt Generator.

FIG. 2 is a process flow diagram illustrating the unified execution pipeline from transaction proposal through compliance check, voice authentication, PQC signature generation, and execution transmission.

FIG. 3 is a data structure diagram illustrating the seven-tier asset compliance classification system and the quarantine state machine.

LIST OF FIGURE ELEMENTS

FIG. 1 Elements:

- 310: Transaction Proposal Input
- 320: Sharia Compliance Engine
- 321: Sharia Compliance Database
- 322: Real-Time Compliance Checker Module
- 323: Asset Quarantine Module
- 324: Periodic Re-Screening Module
- 325: Compliance Audit Trail Module
- 330: Voice Biometric Authentication Module
- 331: Voice Sample Input Interface
- 332: Feature Extraction Engine (spectral + temporal)
- 333: Voice Signature Database (encrypted biometric vault)
- 334: Similarity Score Computation Engine
- 335: Authentication Decision Gate
- 340: Post-Quantum Signature Module
- 341: Dilithium-3 (ML-DSA-65) Signature Engine
- 342: Kyber-768 (ML-KEM-768) Session Key Module
- 343: Transaction Payload Assembler
- 344: PQC-Signed Transaction Output
- 350: Audit Receipt Generator
- 351: Compliance Receipt Record
- 352: Authentication Receipt Record
- 353: Signature Receipt Record
- 354: Combined PQC-Sealed Audit Receipt
- 360: Execution Transmission Interface
- 370: Governance Pipeline Integration (to OMNIX-PAT-2026-001)

FIG. 2 Elements:

- Start: Transaction Proposed
- Step 410: Asset classification lookup in Sharia Compliance Database
- Decision 420: Asset halal / quarantined / haram?
- Step 430: Transaction parameter compliance check
- Decision 440: Parameters compliant?
- Step 450: Voice sample received
- Step 460: Feature extraction (spectral + temporal)

- Step 470: Similarity score computation against stored signature
- Decision 480: Similarity score $\geq$ threshold?
- Step 490: Transaction payload assembly
- Step 500: Dilithium-3 signature generation
- Step 510: Audit receipt generation and sealing
- Step 520: Execution transmission
- Block nodes: COMPLIANCE BLOCK, QUARANTINE BLOCK, AUTH BLOCK

FIG. 3 Elements:

- Tier 1 through Tier 7: Seven compliance classification tiers
- Quarantine State Machine: UNCLASSIFIED $\rightarrow$ UNDER_REVIEW $\rightarrow$ CLASSIFIED (Tier 1–7) $\rightarrow$ QUARANTINED $\rightarrow$ RESOLVED
- Screening cycle arrows: Quarterly re-screening trigger

DETAILED DESCRIPTION OF THE PREFERRED EMBODIMENTS

I. SYSTEM OVERVIEW AND UNIFIED PIPELINE

Referring to FIG. 1 and FIG. 2, the Ethical and Quantum-Secure Execution Framework (hereinafter "the Framework") operates as a pre-execution enforcement layer that intercepts every proposed transaction before it reaches the execution environment. The Framework enforces the following invariant: no transaction is transmitted to the execution environment unless (1) the asset and transaction parameters have been verified as Sharia-compliant, and (2) the transaction has been authorized by a verified user identity and cryptographically signed with a post-quantum digital signature.

The Framework is designed for deployment in automated financial systems, specifically digital asset trading platforms serving Islamic finance market participants. It may additionally be deployed as a standalone authorization layer for any financial system requiring ethical compliance enforcement combined with quantum-resistant identity authorization.

The Framework connects to the governance checkpoint pipeline of OMNIX-PAT-2026-001 through a Pipeline Integration Interface (370). In this integrated configuration, the compliance status and authentication status of a proposed transaction contribute signals to the governance checkpoint evaluation, and any BLOCK verdict from the governance pipeline additionally prevents execution regardless of compliance or authentication status.

II. SHARIA COMPLIANCE ENGINE (320)

II.A. Sharia Compliance Database (321)

The Sharia Compliance Database is a structured data store containing compliance classification records for a plurality of digital assets. Each record in the database comprises:

- **asset_identifier:** A unique identifier for the digital asset (e.g., ticker symbol, contract address, ISIN equivalent).
- **asset_name:** The human-readable name of the digital asset.
- **compliance_tier:** The compliance classification tier assigned to the asset (Tier 1 through Tier 7, as described below).
- **classification_rationale:** A structured record of the Islamic jurisprudence criteria and scholarly sources supporting the classification.
- **classification_date:** The date on which the current classification was assigned.
- **classification_authority:** The scholarly source or supervisory authority responsible for the classification.
- **last_screening_date:** The date on which the asset was most recently re-screened.
- **next_scheduled_screening:** The date on which the next periodic re-screening is scheduled.
- **quarantine_status:** Boolean flag indicating whether the asset is currently quarantined.

- **quarantine_reason:** If quarantined, the reason for the quarantine status.

II.B. Seven-Tier Compliance Classification System

Referring to FIG. 3, the Sharia Compliance Database classifies digital assets into a seven-tier system:

Tier	Classification	Trading Status	Description
Tier 1	Fully Compliant — Halal	Permitted	Asset fully conforms to Sharia principles; no interest, no excessive speculation, no prohibited industry exposure
Tier 2	Compliant with Conditions — Halal with Monitoring	Permitted with alerts	Asset is compliant but has characteristics requiring ongoing monitoring (e.g., minor permissible uncertainty in protocol governance)
Tier 3	Disputed — Scholarly Review Required	Permitted with restriction	Significant scholarly disagreement; trading permitted at reduced position limits pending resolution
Tier 4	Under Active Review — Quarantine Candidate	Restricted pending review	Recent development (e.g., protocol upgrade) has raised compliance concerns; asset placed under active review
Tier 5	Conditionally Prohibited	Restricted	Asset violates specific Sharia criteria but may be permissible for specific transaction types (e.g., spot only, no leverage)
Tier 6	Predominantly Prohibited — Haram	Blocked	Asset predominantly violates Sharia principles; trading blocked except for approved unwinding of existing positions
Tier 7	Fully Prohibited — Haram	Blocked	Asset fully violates Sharia principles (e.g., explicit interest-bearing mechanics, pure speculation instrument); all trading blocked

II.C. Islamic Jurisprudence Criteria

The compliance classification of each digital asset is evaluated against the following Islamic jurisprudence criteria, derived from and citing scholarly sources including standards published by the Accounting and Auditing Organization for Islamic Financial Institutions (AAOIFI) and guidance from recognized Islamic finance scholars including Mufti Muhammad Taqi Usmani:

Criterion 1 — Riba (Interest) Prohibition:

The asset must not incorporate interest-bearing mechanics. For digital assets, this criterion evaluates: (a) whether the asset's yield or return mechanism is interest-based (e.g., staking rewards structured as a fixed interest rate); (b) whether the asset represents a claim on an interest-bearing instrument; and (c) whether the underlying protocol generates income through interest-based lending.

Criterion 2 — Gharar (Excessive Uncertainty) Prohibition:

The asset must not involve excessive uncertainty in contract terms, delivery, or value. For digital assets, this criterion evaluates: (a) whether the asset's value is sufficiently deterministic to constitute a valid contract object under Islamic law; (b) whether the smart contract governing the asset contains ambiguous or indeterminate terms; and (c) whether the asset's price is subject to artificial manipulation mechanisms.

Criterion 3 — Maysir (Gambling/Speculation) Prohibition:

The asset must not be a pure speculation instrument. For digital assets, this criterion evaluates: (a) whether the asset has a productive economic use beyond speculation; (b) whether the asset's return is based on productive activity or purely on price movement; and (c) whether the asset is structured as a derivative with no underlying productive asset.

II.D. Real-Time Compliance Checker Module (322)

The Real-Time Compliance Checker Module evaluates each proposed transaction against the Sharia Compliance Database before execution. For each proposed transaction, the module performs the following checks in sequence:

Check 1 — Asset Classification Lookup: The asset identifier is looked up in the Sharia Compliance Database. If the asset is not present in the database, the asset is treated as UNCLASSIFIED and the transaction is blocked pending classification.

Check 2 — Compliance Tier Evaluation: If the asset is classified as Tier 6 or Tier 7, the transaction is immediately blocked. If the asset is classified as Tier 5, the transaction is evaluated against the specific permitted transaction types for Tier 5 assets.

Check 3 — Quarantine Status Check: If the asset's quarantine_status flag is set to true, the transaction is blocked regardless of the compliance tier classification.

Check 4 — Transaction Parameter Compliance: The transaction parameters (leverage ratio, position duration, derivative structure, yield mechanism) are evaluated against Sharia criteria. Transactions with interest-bearing components, excessive leverage, or derivative structures not backed by a productive underlying asset are blocked.

II.E. Asset Quarantine Module (323)

The Asset Quarantine Module maintains the quarantine status of digital assets and enforces quarantine blocks. A quarantine event may be triggered by:

- **Automatic triggering:** The Real-Time Compliance Checker detects a transaction characteristic that raises a compliance concern not reflected in the current tier classification (e.g., a protocol upgrade announcement that may introduce interest-bearing mechanics).
- **Periodic re-screening trigger:** The Periodic Re-Screening Module (324) identifies a change in the asset's characteristics during a scheduled re-screening cycle.
- **External alert trigger:** An authorized compliance authority flags the asset for immediate quarantine review.

When a quarantine event is triggered, the Asset Quarantine Module:

- Sets the quarantine_status flag to true in the Sharia Compliance Database
- Logs a quarantine event record with the trigger reason, timestamp, and asset identifier
- Notifies the designated compliance authority for review
- Prevents all transactions involving the quarantined asset until the quarantine is resolved

Quarantine resolution requires explicit action by a designated compliance authority: the quarantine cannot be resolved automatically by the system.

II.F. Periodic Re-Screening Module (324)

The Periodic Re-Screening Module automatically initiates re-evaluation of the compliance classification of all digital assets in the Sharia Compliance Database at configurable intervals. In the preferred embodiment, re-screening occurs at minimum on a quarterly basis. Re-screening may be triggered more frequently for Tier 3, Tier 4, and quarantined assets.

The re-screening process evaluates whether any material changes in the asset's protocol, governance structure, usage patterns, or underlying economics have occurred since the previous screening that would warrant a change in compliance tier classification.

III. VOICE BIOMETRIC AUTHENTICATION MODULE (330)

The Voice Biometric Authentication Module provides the first layer of the dual-layer authorization system. It verifies the identity of the user requesting transaction authorization by comparing a voice sample provided by the user against a stored voice signature.

III.A. Voice Sample Input Interface (331)

In a preferred embodiment, the Voice Sample Input Interface accepts voice samples transmitted through a messaging platform bot interface (e.g., a Telegram bot interface), wherein the voice sample is received as a voice message transmitted through the messaging platform. Alternative embodiments support voice input through a dedicated mobile application microphone interface, a web application audio API, or a telephony interface.

III.B. Feature Extraction Engine (332)

The Feature Extraction Engine extracts a set of voice biometric features from the received voice sample. In a preferred embodiment, the extracted features comprise:

Spectral Features: Mel-Frequency Cepstral Coefficients (MFCCs) characterizing the frequency-domain characteristics of the voice sample. MFCCs capture the shape of the vocal tract and are robust to background noise.

Temporal Features: The duration and temporal envelope of the voice sample, capturing speaking rate and rhythm characteristics.

The combined feature set characterizes the biometric identity of the speaker in a compact numerical representation.

III.C. Similarity Score Computation (334)

The Similarity Score Computation Engine compares the extracted feature set against the stored voice signature associated with the requesting user. In one embodiment, the similarity score is computed as a weighted combination of two feature components:

$$\text{similarity_score} = 0.70 \times \text{spectral_match} + 0.30 \times \text{temporal_match}$$

In other embodiments, the weighting between spectral and temporal components may be configured to any values appropriate to the deployment context, the characteristics of the enrolled user population, and the acoustic environment. Alternative embodiments may incorporate additional biometric feature dimensions (e.g., pitch envelope, formant trajectories, rhythm patterns) with corresponding configurable weights. The structural requirement — that the similarity score be derived from a plurality of independent acoustic feature dimensions combined into a single scalar score for comparison against a threshold — is preserved across all embodiments regardless of the specific weighting scheme.

where:

- **spectral_match** is the cosine similarity between the extracted MFCC feature vector and the stored voice signature MFCC vector, normalized to [0.0, 1.0]
- **temporal_match** is a normalized measure of the similarity between the duration and temporal envelope of the received voice sample and the stored voice signature

A similarity score of 1.0 indicates a perfect match; a similarity score of 0.0 indicates maximum dissimilarity.

III.D. Authentication Decision Gate (335)

The Authentication Decision Gate evaluates the similarity score against a configurable authentication threshold. In one embodiment, the threshold is set at 0.75, meaning that a similarity score of at least 0.75 is required for authentication to succeed. In other embodiments, this threshold may be set to any value appropriate to the security requirements of the deployment context, the sensitivity of the transaction types being authorized, and the biometric characteristics of the enrolled user population. The structural requirement — that authentication must succeed before the PQC signature layer is invoked — is preserved across all embodiments regardless of the specific threshold value. If the similarity score falls below the configured threshold, the authentication attempt is rejected and the proposed transaction is blocked.

In a preferred embodiment, the Voice Signature Database (333) stores voice signatures in an encrypted biometric vault using post-quantum encryption. The system additionally requires periodic re-enrollment of voice signatures at predetermined intervals (e.g., every 90 days) to account for natural voice changes over time.

IV. POST-QUANTUM SIGNATURE MODULE (340)

The Post-Quantum Signature Module provides the second layer of the dual-layer authorization system. Upon successful voice biometric authentication, it generates a post-quantum digital signature over the complete transaction payload.

IV.A. Session Key Establishment (342)

Prior to signature generation, a secure communication session is established between the user's device and the transaction processing server using Kyber-768 (ML-KEM-768) as specified in NIST FIPS 203. Kyber-768 is a module-lattice-based key encapsulation mechanism that provides quantum-resistant establishment of a shared session key without exchanging the key over the communication channel.

The quantum-resistant session establishment ensures that session keys cannot be compromised even by an adversary with access to a quantum computer, protecting the confidentiality and integrity of the transaction authorization process.

IV.B. Transaction Payload Assembly (343)

The Transaction Payload Assembler assembles the complete transaction payload to be signed, comprising:

- Transaction identifier (UUID)
- Timestamp (UTC, millisecond resolution)
- Asset identifier
- Transaction type (BUY, SELL, TRANSFER, etc.)
- Transaction parameters (quantity, price, leverage ratio, execution conditions)
- User identifier (derived from the voice authentication process)
- Compliance clearance reference (from the Sharia Compliance Engine)
- Nonce (cryptographic random value preventing replay attacks)

IV.C. Dilithium-3 Signature Generation (341)

The Dilithium-3 (ML-DSA-65) Signature Engine, implementing the algorithm specified in NIST FIPS 204, generates a digital signature over the assembled transaction payload. Dilithium-3 provides:

- Security level equivalent to at least 128-bit classical security
- Resistance to known quantum computing attacks, including Shor's algorithm and Grover's algorithm
- Compact signature size suitable for real-time transaction processing
- Deterministic signature generation (same private key + same message → same signature), enabling audit verification

The private signing key is stored in a secure hardware environment (preferred: hardware security module or equivalent). The corresponding public verification key is made available to authorized audit parties for signature verification.

IV.D. Signed Transaction Output (344)

The output of the Post-Quantum Signature Module is a PQC-signed transaction record comprising the assembled transaction payload and the Dilithium-3 digital signature. This record is transmitted to the execution environment and persisted in the audit log.

V. AUDIT RECEIPT GENERATOR (350)

The Audit Receipt Generator assembles a comprehensive audit receipt for each processed transaction, comprising three component records sealed into a single cryptographically protected document:

Compliance Receipt Record (351):

- Asset identifier and compliance tier at time of transaction
- Transaction parameter compliance evaluation results
- Quarantine status at time of transaction
- Compliance Engine version and database version identifiers
- Timestamp of compliance evaluation

Authentication Receipt Record (352):

- User identifier
- Authentication method (voice biometric)
- Similarity score (above threshold — exact score may be omitted for privacy)
- Authentication timestamp
- Voice signature re-enrollment status

Signature Receipt Record (353):

- Transaction payload hash
- Dilithium-3 signature
- Public key reference for verification
- Signature generation timestamp
- NIST FIPS 204 algorithm identifier

The three component records are assembled into a Combined PQC-Sealed Audit Receipt (354) and signed with a second Dilithium-3 signature using the system's audit key, providing tamper-evident evidence of the complete transaction authorization process.

VI. UNIFIED EXECUTION PIPELINE — PROCESS FLOW

Referring to FIG. 2, the complete unified execution pipeline processes each proposed transaction as follows:



Every exit path that does not reach Step 520 generates a BLOCK audit record documenting the reason for the block, the stage at which the block was issued, and the timestamp. No transaction reaches the execution environment without passing all stages.

ALTERNATIVE EMBODIMENTS

Embodiment A: Multi-Madhhab Compliance Framework

In an alternative embodiment, the Sharia Compliance Engine maintains separate compliance classification databases for different schools of Islamic jurisprudence (madhhabs), including Hanafi, Maliki, Shafi'i, and Hanbali. Users may configure their preferred madhhab, and the compliance evaluation applies the corresponding classification database. This embodiment accommodates the scholarly diversity within Islamic finance.

Embodiment B: Fingerprint Biometric Alternative

In an alternative embodiment, the voice biometric authentication is supplemented by or replaced with fingerprint biometric authentication, using a biometric sensor integrated with the user's device. The feature extraction and similarity score computation processes are adapted for fingerprint data. The dual-layer structure (biometric + PQC signature) is preserved.

Embodiment C: Multi-Signatory Authorization

In an alternative embodiment, the dual-layer authorization is extended to require multiple authorized signatories for transactions exceeding a configurable monetary threshold. Each signatory provides independent voice biometric authentication, and the transaction payload includes all signatories' Dilithium-3 signatures in a multi-signature structure.

Embodiment D: Non-Islamic Ethical Compliance Domains

In an alternative embodiment, the Sharia Compliance Engine is generalized to an Ethical Compliance Engine that enforces different ethical frameworks as configured by the deploying institution. Alternative ethical frameworks may include ESG (Environmental, Social, Governance) compliance, UNPRI (UN Principles for Responsible Investment) compliance, or domain-specific ethical restrictions (e.g., prohibition on transactions in specified industry sectors). The asset quarantine mechanism, compliance tier classification, and real-time checker architecture remain structurally identical across ethical framework configurations.

Embodiment F: Dynamic Compliance Tier Count

The system may operate with any number of compliance tiers, from a minimal binary classification (halal / haram) to the preferred seven-tier system or a more granular structure. The asset quarantine mechanism and the real-time compliance checker operate identically regardless of the number of tiers, provided that each tier maps to a defined set of transaction restrictions.

Embodiment G: Alternative Ethical Framework Configuration

The Sharia Compliance Engine may be configured to enforce alternative ethical frameworks selected by the deploying institution, including ESG compliance, UNPRI compliance, or domain-specific ethical restrictions. The seven-tier classification structure, asset quarantine mechanism, real-time compliance checker, periodic re-screening module, and compliance audit trail remain structurally identical across ethical framework configurations. The system may support simultaneous enforcement of multiple ethical frameworks in a layered configuration.

Embodiment H: Alternative Post-Quantum Algorithms

The system may operate with alternative post-quantum digital signature algorithms standardized by NIST or other recognized standards bodies, in addition to or in replacement of Dilithium-3 (ML-DSA-65, NIST FIPS 204). The dual-layer authorization structure — biometric first layer, PQC signature second layer — is preserved regardless of the specific post-quantum algorithm selected.

Embodiment E: Hardware Security Module Integration

In an alternative embodiment, the Dilithium-3 private signing key is stored in and all signature operations are performed within a FIPS 140-3 validated hardware security module (HSM), providing physical protection against key extraction attacks in addition to the quantum-resistance of the signature algorithm.

EXAMPLES OF USE

Example 1: Halal Transaction — Full Pipeline Execution

A user submits a request to purchase 0.5 BTC through an automated trading system. The Sharia Compliance Engine looks up BTC in the database and finds it classified as Tier 2 (Compliant with Conditions — spot trading only, no interest-bearing lending). The transaction parameters (spot purchase, no leverage) are consistent with Tier 2 conditions. The system proceeds to voice authentication. The user submits a voice message through the Telegram bot interface. The Feature Extraction Engine computes a similarity score of 0.84, exceeding the 0.75 threshold. Authentication succeeds. The

Transaction Payload Assembler creates the payload; the Dilithium-3 engine generates the signature. The Audit Receipt Generator seals the compliance record, authentication record, and signature record into a combined PQC-sealed receipt. The signed transaction is transmitted to the exchange.

Example 2: Compliance Block — Quarantined Asset

A user submits a request to purchase an altcoin that has been placed in quarantine following a protocol upgrade that introduced a staking yield mechanism with fixed interest rate characteristics. The Sharia Compliance Engine finds the asset's quarantine_status flag set to true. The transaction is blocked at Step 420 without proceeding to voice authentication. A COMPLIANCE BLOCK audit record is generated, documenting the quarantine reason and timestamp.

Example 3: Authentication Block — Similarity Below Threshold

A user submits a voice sample for a large transaction. The Feature Extraction Engine computes spectral features; the similarity computation yields a score of 0.68 against the stored voice signature, below the 0.75 threshold. The Authentication Decision Gate rejects the authentication. The transaction is blocked at Step 480. An AUTH BLOCK audit record is generated. The user may attempt re-authentication after a configurable lockout period.

Example 4: Integration with Governance Pipeline

A proposed transaction clears both the Sharia Compliance Engine and the voice biometric authentication successfully. The signed transaction is submitted to the governance checkpoint pipeline (OMNIX-PAT-2026-001). At CP-6, the Decision Contradiction Index evaluates the regime signals and finds a DCI score of 76 (CONTRADICTIONARY). The governance pipeline issues a BLOCK verdict. The transaction does not reach the execution environment despite having passed both ethical compliance and identity authorization. This illustrates the complementary relationship between the Ethical and Quantum-Secure Execution Framework and the Governance Control Architecture.

TECHNICAL ADVANTAGES

- **Real-time pre-execution enforcement:** The compliance engine and authorization system operate at the execution boundary, blocking non-compliant or unauthorized transactions before they reach the execution environment. This provides enforcement at the critical point where a proposed transaction becomes irreversible.
- **Asset quarantine as independent enforcement mechanism:** The quarantine mechanism operates independently of the per-transaction compliance checker, providing a categorical block on all transactions involving a quarantined asset without requiring per-transaction analysis.
- **Quantum-resistant authorization chain:** The use of Dilithium-3 (NIST FIPS 204) and Kyber-768 (NIST FIPS 203) provides an authorization chain that remains cryptographically secure against both current and anticipated quantum computing attacks.
- **First known combination of voice biometrics and PQC in financial authorization:** To the inventor's knowledge, this is the first system to combine voice biometric authentication with NIST FIPS 204-compliant post-quantum digital signatures for financial transaction authorization.
- **Complete, tamper-evident audit trail:** The combined PQC-sealed audit receipt provides a complete record of compliance evaluation, identity verification, and authorization for every transaction, suitable for regulatory examination.
- **Unified pipeline enforces both ethical and security requirements:** By connecting compliance enforcement and identity authorization in a single sequential pipeline, the Framework ensures that both requirements must be satisfied independently before execution proceeds. Neither can substitute for the other.

CLAIMS

(Note: Claims are included for reference. Formal claims will be refined in the corresponding nonprovisional application.)

Claim 1 (Broad — Compliance System) — A computer-implemented system for enforcing Islamic financial compliance in digital asset trading, comprising: a Sharia compliance database classifying digital assets according to Islamic

jurisprudence criteria; a real-time compliance checker that evaluates proposed transactions against said database prior to execution; and an asset quarantine module that blocks all transactions involving quarantined assets until compliance status is resolved.

Claim 2 (Specific — Seven-Tier Classification) — The system of Claim 1, wherein said Sharia compliance database classifies digital assets into a seven-tier classification system, with Tier 1 as fully compliant and Tier 7 as fully prohibited, each tier having associated transaction restrictions.

Claim 3 (Specific — Jurisprudence Criteria) — The system of Claim 1, wherein said Islamic jurisprudence criteria comprise prohibition of riba (interest), prohibition of gharar (excessive uncertainty), and prohibition of maysir (speculation), evaluated against the characteristics of each digital asset.

Claim 4 (Specific — Periodic Re-Screening) — The system of Claim 1, further comprising a periodic re-screening module that automatically re-evaluates compliance classifications at configurable intervals, wherein re-screening triggers asset quarantine when material compliance-relevant changes are detected.

Claim 5 (Broad — Dual Authorization) — A computer-implemented method for authorizing financial transactions using dual-layer authentication, comprising: receiving a voice sample; extracting biometric features from said voice sample; comparing said features against a stored voice signature to generate a similarity score; upon said similarity score exceeding a threshold, generating a post-quantum digital signature over the transaction payload using an algorithm compliant with NIST FIPS 204; and transmitting the signed transaction for execution.

Claim 6 (Specific — Dilithium-3) — The method of Claim 5, wherein said post-quantum digital signature is generated using Dilithium-3 (ML-DSA-65) as specified in NIST FIPS 204, providing security equivalent to at least 128-bit classical security and resistance to quantum computing attacks.

Claim 7 (Specific — Session Key) — The method of Claim 5, further comprising establishing a secure session between user device and server using Kyber-768 (ML-KEM-768) as specified in NIST FIPS 203.

Claim 8 (Specific — Similarity Score Weighting) — The method of Claim 5, wherein, in one embodiment, the similarity score is computed as $0.70 \times \text{spectral feature match} + 0.30 \times \text{temporal feature match}$; and wherein in other embodiments the weighting coefficients are configurable to any values appropriate to the deployment context, provided that the similarity score is derived from a plurality of independent acoustic feature dimensions combined into a single scalar value for threshold comparison.

Claim 9 (Broad — Unified Pipeline) — A computer-implemented system enforcing ethical compliance and quantum-resistant identity authorization in a single sequential pre-execution pipeline, wherein compliance clearance and biometric authorization are each independently required for execution, and wherein failure of either independently blocks the proposed transaction.

Claim 10 (Broad — Audit Receipt) — The system of Claim 9, further comprising an audit receipt generator that produces a combined cryptographically sealed audit record comprising a compliance evaluation record, an authentication record, and a post-quantum signature record for every processed transaction.

Claim 11 (Governance Integration) — The system of Claim 9, wherein the output of the unified pipeline is additionally subject to evaluation by an automated decision governance pipeline comprising independent checkpoints each having veto authority, and wherein a block verdict from said governance pipeline prevents execution regardless of compliance clearance and authentication status.

Claim 12 (Broad System — Maximum Coverage) — A system configured to enforce ethical compliance requirements and identity authorization requirements as independent pre-execution conditions for automated financial transactions, the system comprising one or more processors and a non-transitory computer-readable medium storing instructions that cause the system to: evaluate a proposed transaction against a stored classification database to determine whether the transaction satisfies applicable ethical requirements; independently verify the identity of the requesting party through biometric analysis; and prevent execution of said transaction unless both the ethical compliance evaluation and the identity verification independently succeed, wherein failure of either independently blocks execution without regard to the

result of the other.

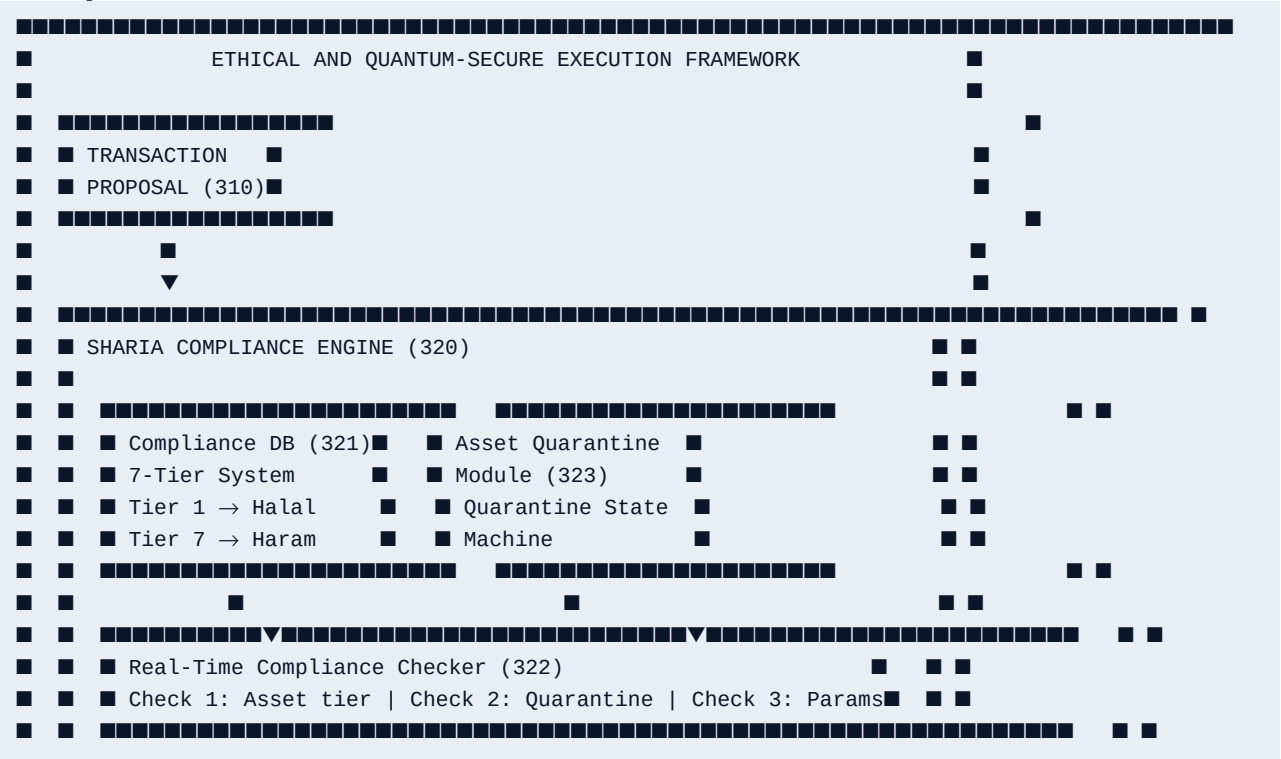
Claim 13 (Broad Method — Maximum Coverage) — A computer-implemented method for authorizing automated financial transactions through dual independent pre-execution requirements, comprising: evaluating a proposed transaction against ethical compliance criteria to produce a compliance determination; evaluating the identity of the requesting party through biometric feature comparison to produce an authentication determination; generating a quantum-resistant cryptographic authorization record upon successful completion of both evaluations; and transmitting said transaction for execution only upon generation of said authorization record, wherein neither a successful compliance determination alone nor a successful authentication determination alone is sufficient to authorize execution.

ABSTRACT

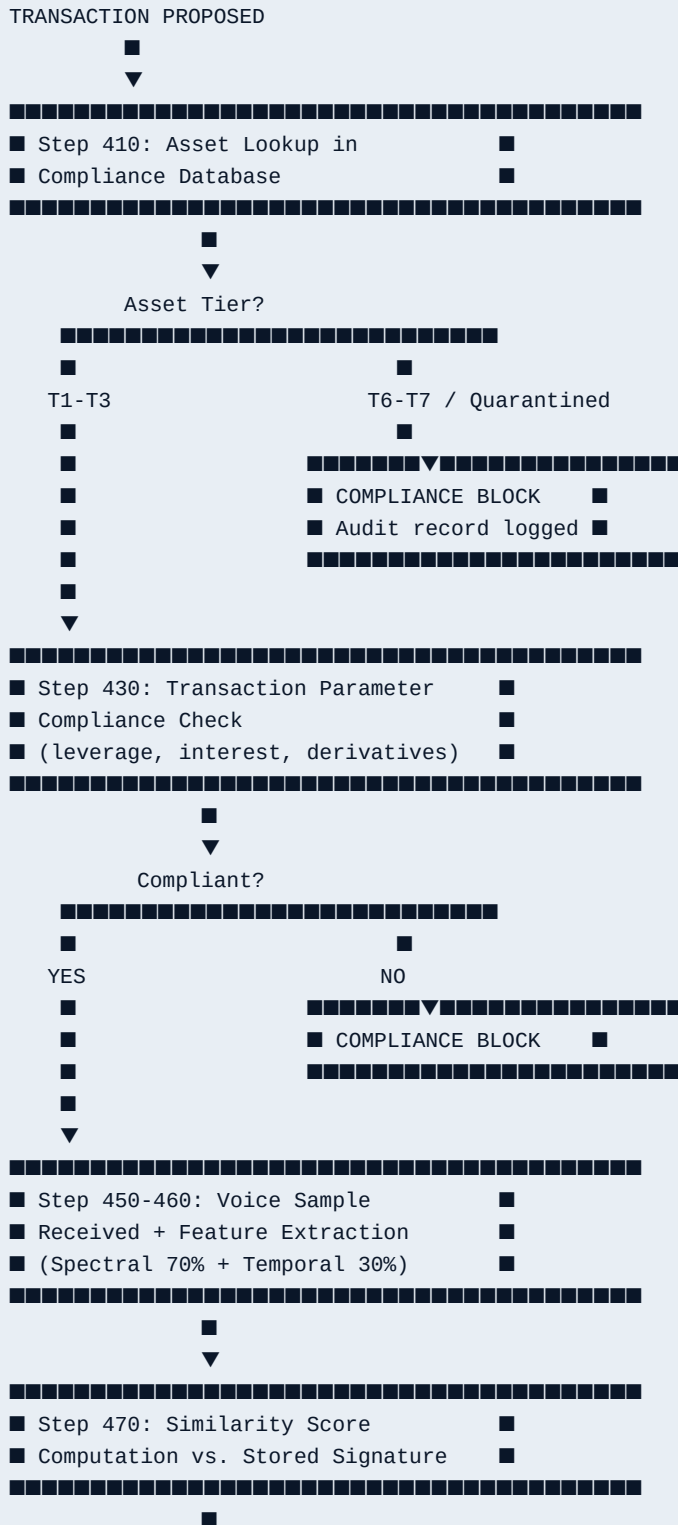
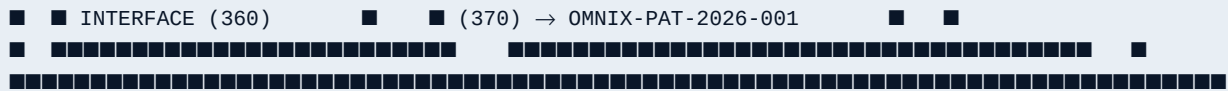
A computer-implemented Ethical and Quantum-Secure Execution Framework provides pre-execution enforcement of two independent requirements for digital asset transactions. An Automated Sharia Compliance Engine evaluates proposed transactions against a seven-tier compliance database derived from Islamic jurisprudence criteria — *riba*, *gharar*, and *maysir* prohibitions — and enforces compliance through an asset quarantine mechanism that categorically blocks all transactions involving quarantined assets pending compliance resolution. A Dual-Layer Transaction Authorization system requires sequential voice biometric authentication (similarity score against stored voice signature; in one embodiment: 70% spectral + 30% temporal weighting, configurable in other embodiments) and post-quantum digital signature generation using Dilithium-3 (ML-DSA-65, NIST FIPS 204), with session establishment via Kyber-768 (ML-KEM-768, NIST FIPS 203). Both components are connected through a unified sequential execution pipeline in which compliance clearance and identity authorization are each independently required and neither can substitute for the other. A combined post-quantum-sealed audit receipt documents the compliance evaluation, authentication, and signature for every processed transaction. To the inventor's knowledge, this is the first system combining voice biometric authentication with NIST FIPS 204-compliant post-quantum digital signatures for financial transaction authorization.

DRAWINGS

FIG. 1 — System Architecture



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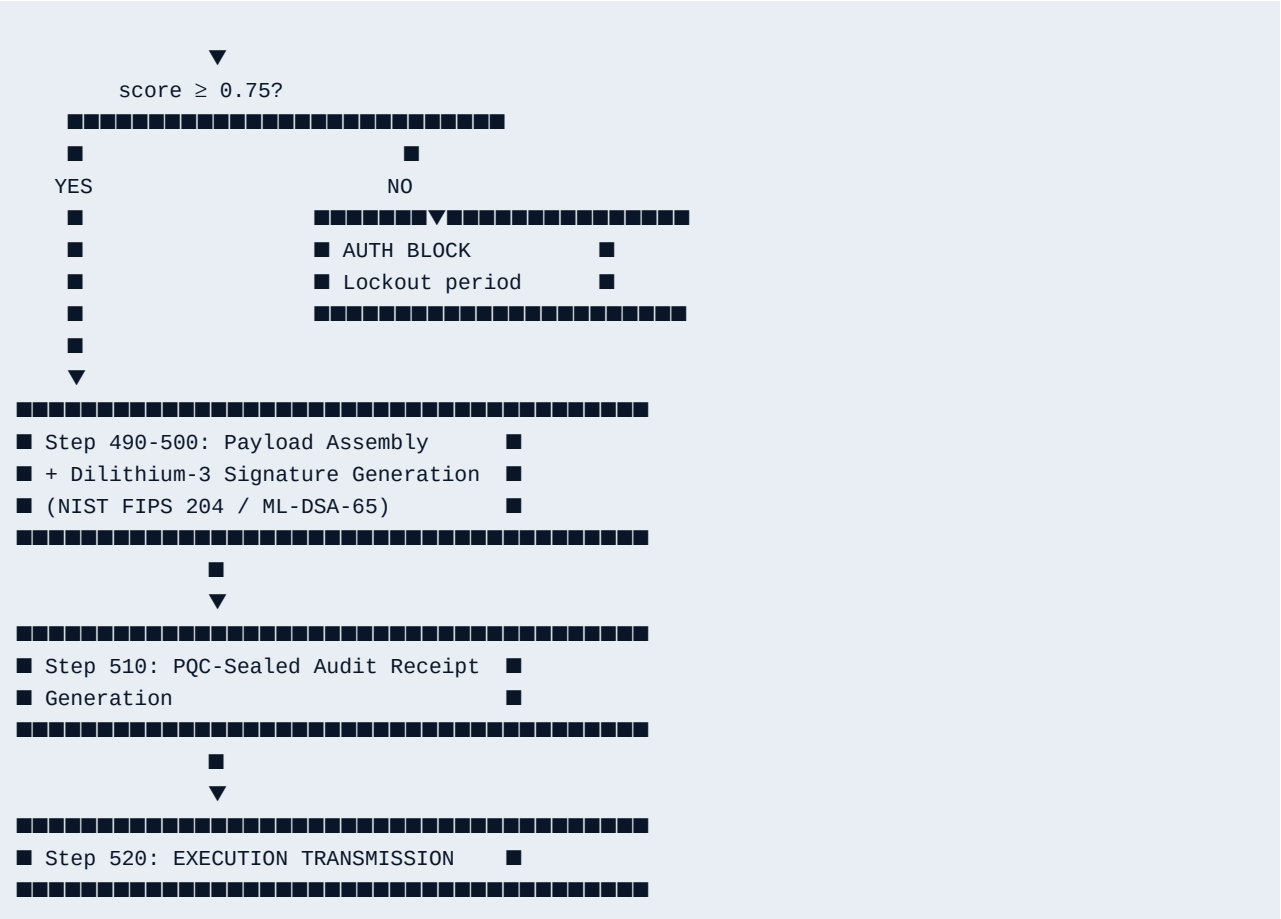
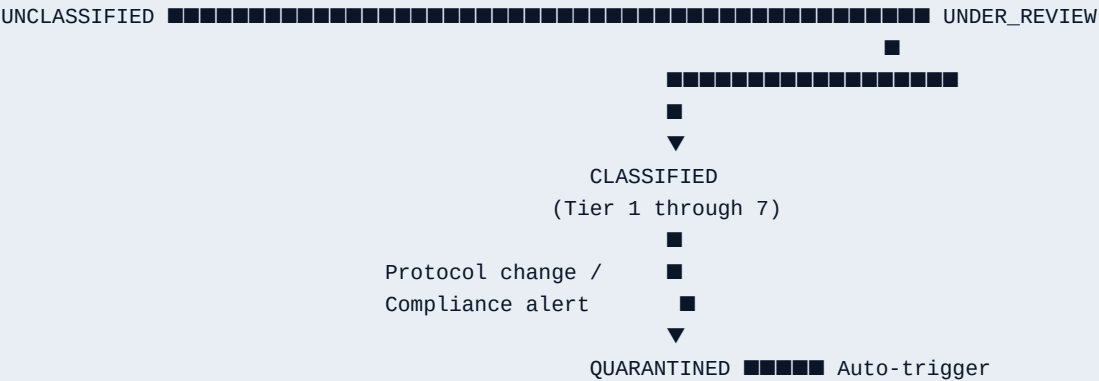


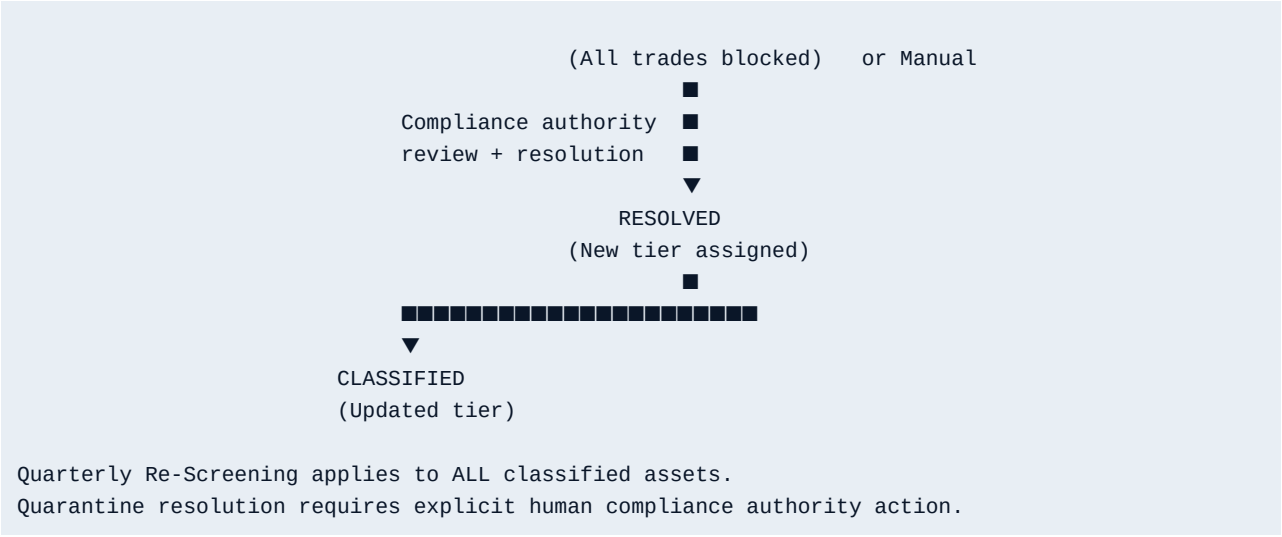
FIG. 3 — Seven-Tier Compliance Classification and Quarantine State Machine

ASSET COMPLIANCE CLASSIFICATION:

Tier 1	FULLY HALAL	→ Trading: PERMITTED
Tier 2	HALAL WITH MONITORING	→ Trading: PERMITTED + ALERTS
Tier 3	SCHOLARLY DISPUTED	→ Trading: RESTRICTED
Tier 4	UNDER REVIEW	→ Trading: RESTRICTED PENDING
Tier 5	COND. PROHIBITED	→ Trading: SPOT ONLY / LIMITED
Tier 6	PREDOMINANTLY HARAM	→ Trading: BLOCKED
Tier 7	FULLY HARAM	→ Trading: FULLY BLOCKED

QUARANTINE STATE MACHINE:





End of Specification — Patent Family 3

Document Reference: OMNIX-PAT-2026-003

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